Shreshth Srivastava

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Work Experience

Network/ML Performance Engineer (Intern), ARM, USA May 2025 – Aug 2025

- Integrated SnortML into ARM's critical Snort3 benchmarking pipeline across **ARM64/x86**, enabling ML-based IDS evaluation and strengthening client engagements in the network security domain.
- Extended internal **Ansible** roles to support SnortML plugin builds, custom rule injection and dynamic benchmarking with **PCAP** inputs.
- Developed **PyTorch** classifiers for HTTP intrusion detection, reducing dependency on handcrafted rules and improving detection granularity via multi-class models (e.g., `generic_sqli`, `time_based`).

Software Engineer, Qualcomm, India Jul 2022 – Aug 2024

- Enhanced GPS accuracy using **EDGNSS** and **RTK**, improving precision tenfold.
- Developed sample Android applications showcasing **Qualcomm SDK** features for **Snapdragon chips**, used as references by leading vendors and OEMs, including **Meta**, **Microsoft**, **Vivo**, and **Xiaomi**.
- Ported applications like **OBS Studio**, **Chromium**, and **GIMP** from **x86** to **ARM64** architecture for Snapdragon platforms, optimizing performance and ensuring compatibility.
- Enhanced **BLE** protocols using **Distributed Context Fabric (DCF)** integrated with **Google Firebase**, enabling low-latency accessory tracking with just a 1% battery impact.
- Customized Android Open Source Project **(AOSP)** builds by integrating cherry-picked features, enabling tailored Android experiences for different hardware platforms and use cases.

Solution Delivery Analyst, Deloitte, India Jul 2021 – May 2022

- Built an automation framework for server diagnostics using **JSch**, **CyberArk**, and Shell scripting, enabling secure remote checks and improving monitoring efficiency by 11%.
- Migrated on-premises applications to **AWS** using **IaC** while securing them with **SAML 2.0** and **OAuth 2.0** authentication systems, cutting costs and improving scalability.

Technology Intern, Deloitte, India Jan 2021 – Apr 2021

- Optimized **JVM** tuning and implemented **load balancing algorithms**, achieving a 50% improvement in server response times as measured on **AppDynamics**.

Education

University of Colorado Boulder, Boulder, CO, USA Aug 2024 – Present
Master of Science in Computer Science
Relevant Coursework: Object-Oriented Analysis and Design, Network Systems, Neural Networks, Foundations of Software Engineering
Graduate Teaching Assistant: Principles of Programming Languages (CSCI 3155), Department of Computer Science

Vellore Institute of Technology (VIT), Vellore, India Jun 2017 – Jun 2021
Bachelor of Technology in Electronics and Communication Engineering, GPA: 8.58/10
Relevant Coursework: Data Structures & Algorithms, Operating Systems, Network Security, Digital-Logic and Design

Publications

Decentralization of Identities Using Blockchain
10th IEEE UP Section International Conference on Electrical, Electronics and Computer Engineering (UPCON-2023), `https://doi.org/10.1109/UPCON59197.2023.10434697`

Projects

- **eBPF Kernel Packet Processor**: Developed a real-time UDP packet filtering and logging system using eBPF in the Linux kernel, to enable secure communication and perform stateful packet analysis.
- **3D OpenGL Simulation**: Designed realistic water rendering with Fresnel effects, normal mapping, and a CPU-based particle system.
- **Design Patterns Benchmarking**: Developed an online quiz system in Java to analyze the impact of various design patterns on application performance with benchmarking for performance evaluation.
- **Cova App**: Developed an Android app for real-time COVID-19 news using Newscatcher and NewsData APIs, featuring a chat room for discussions, master-detail layout, advanced notifications, and MVVM architecture with ROOM.

Certifications

- Cryptography and Information Theory - University of Colorado (Coursera)
- Android Developer Nanodegree - Udacity
- Open Source Software Development Methods - The Linux Foundation (Coursera)
- Problem-Solving (Basic & Intermediate) - HackerRank
- The Unix Workbench - Johns Hopkins University (Coursera)

Skills

Languages: Java, Kotlin, C++, Python, SQL, JavaScript, Dart, Bash, Go
Web & Mobile: Android, Flutter, Spring Boot, Node.js, Firebase, REST APIs
DevOps & Tools: Git, Docker, Kubernetes, Jenkins, CI/CD Pipelines, AWS (**EC2**, **S3**, **CloudFormation**),
Databases: MySQL, PostgreSQL, MongoDB, SQLite
Testing Frameworks: JUnit, Mockito, Postman
Other: Blockchain, Multithreading, Data Structures and Algorithms, API Development, Debugging, Agile Development